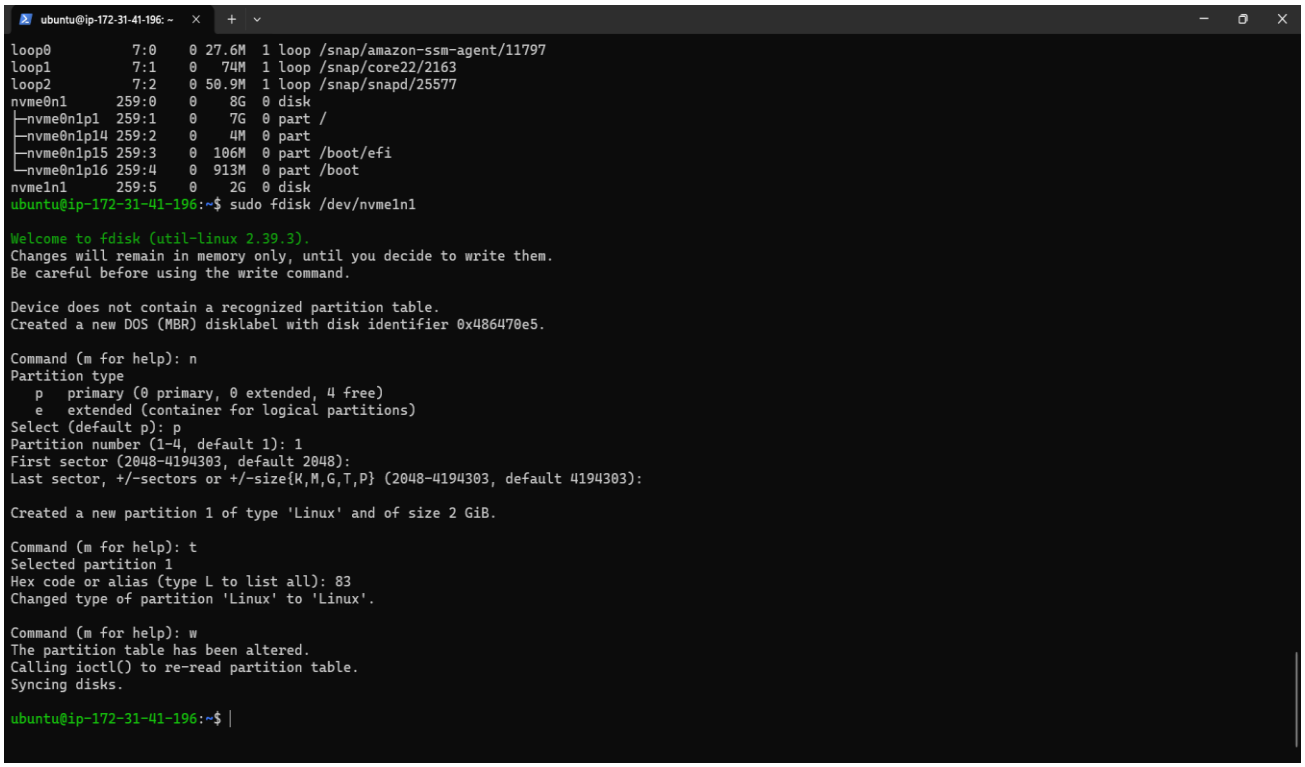
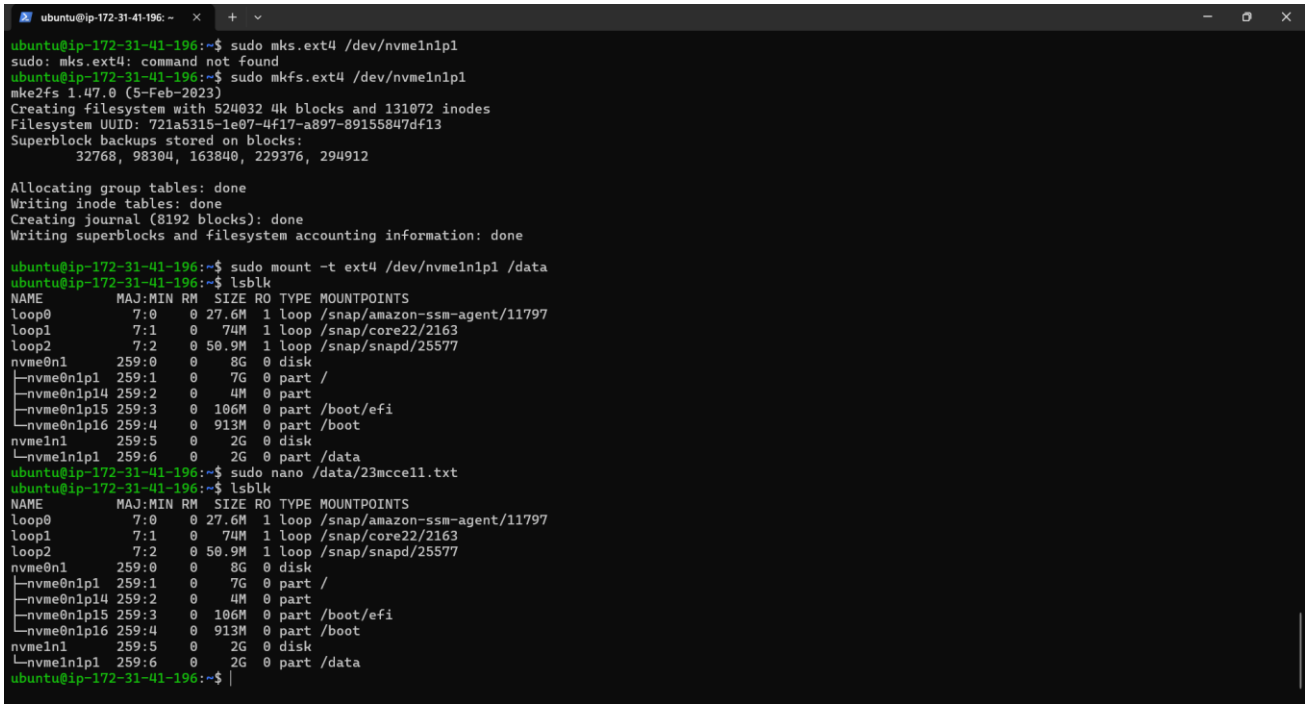
Cloud Computing

Exercise-3 23MCCE11

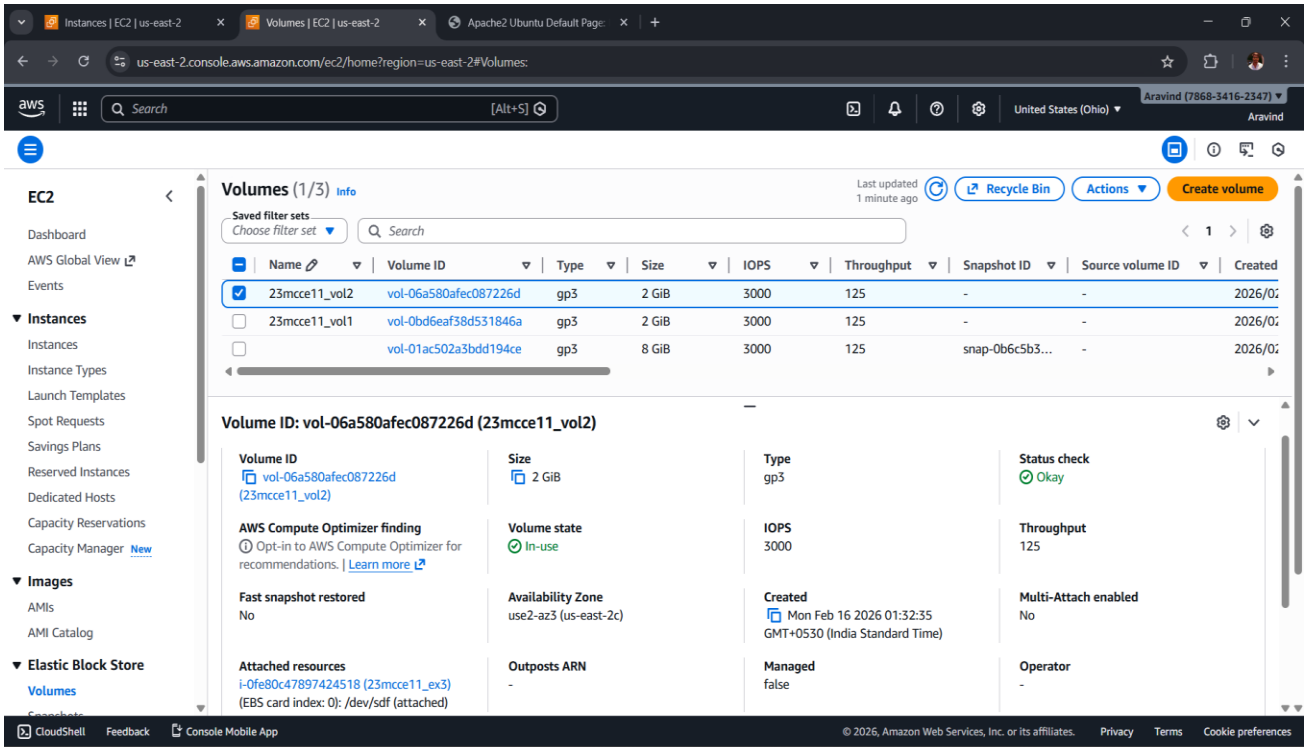
1.

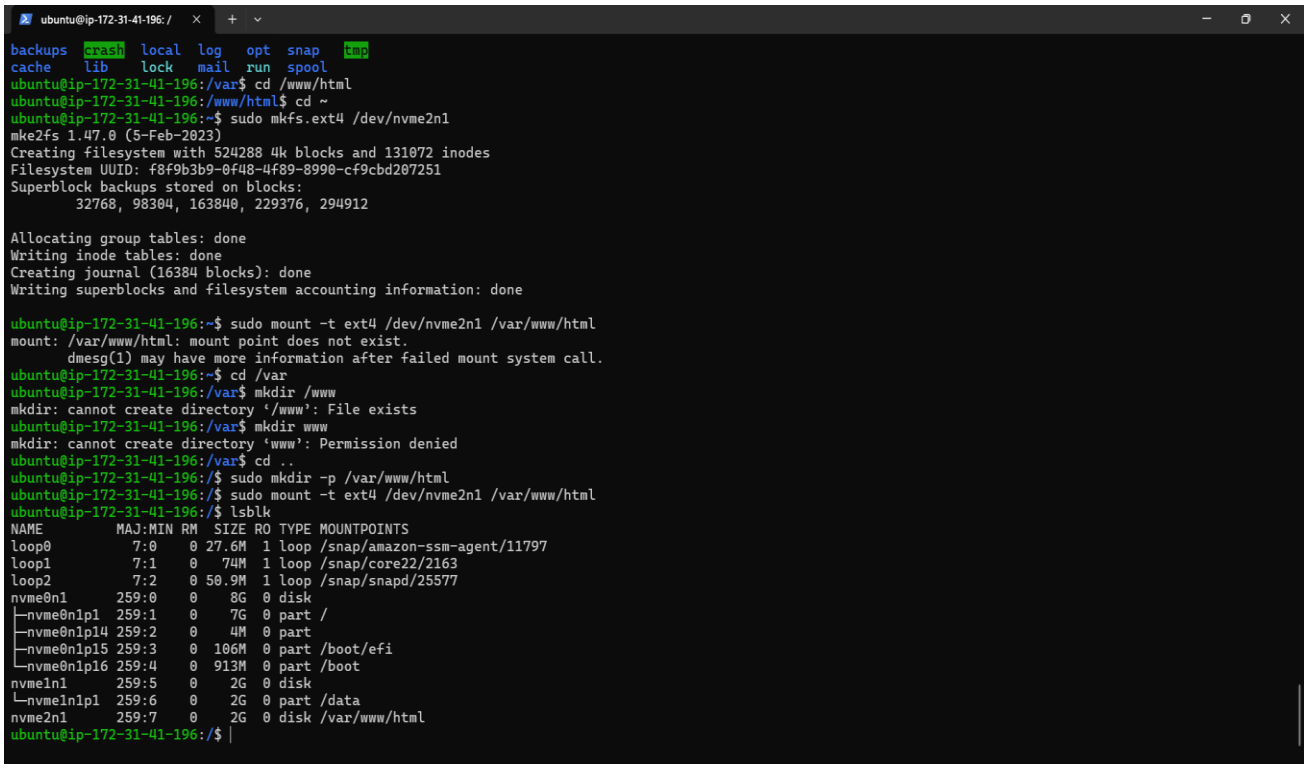




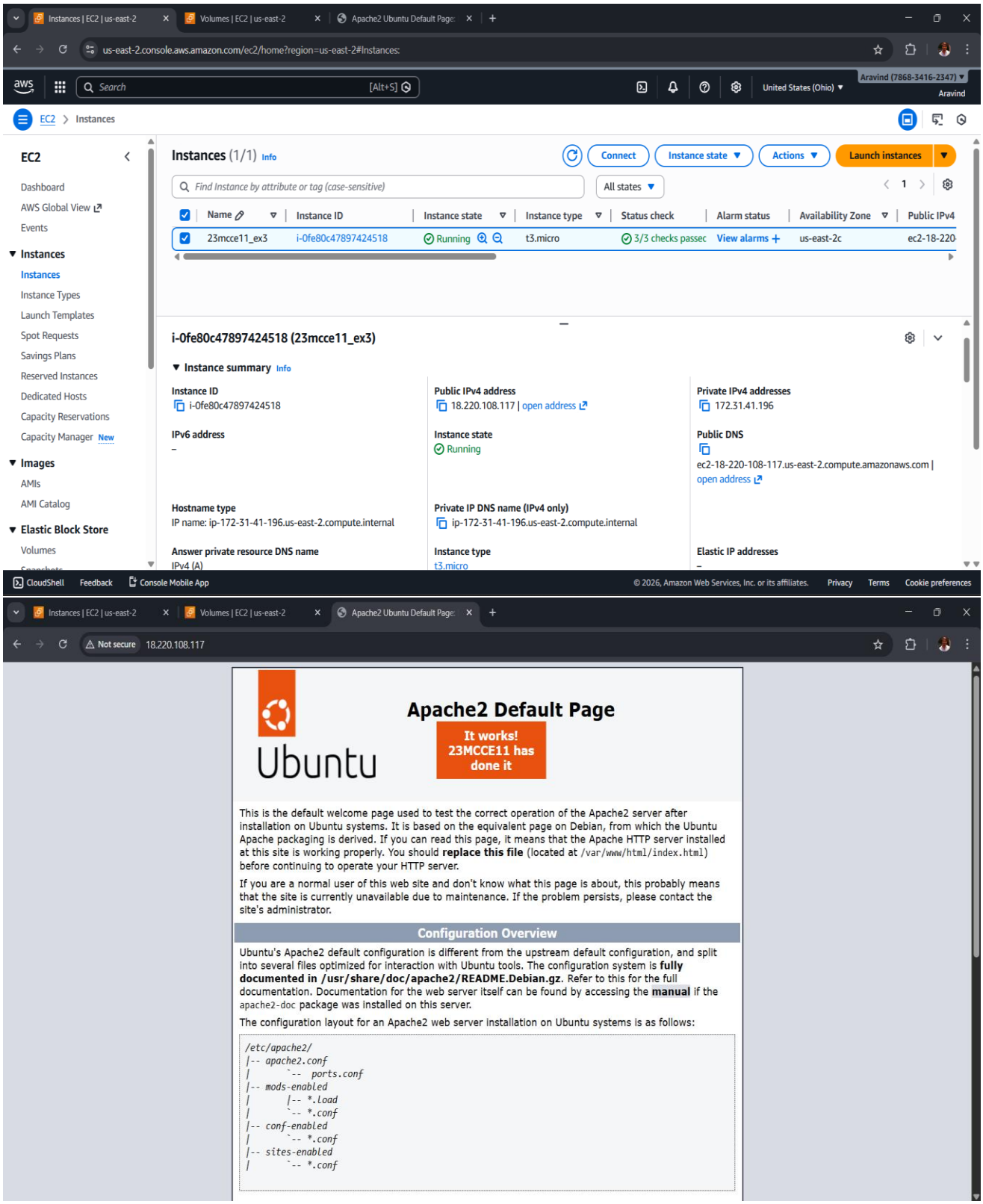
In the 1st image we can see the partition of size 2G named as nvme1n1 (device name - /dev/sdx) and the process of partition. In the 2nd image I used lsblk before creating 23mcce11.txt to see will it make any difference but didn't but with df -h | grep nvme1n1p1 (which I later learned) we can see how much space the partition used.

2.



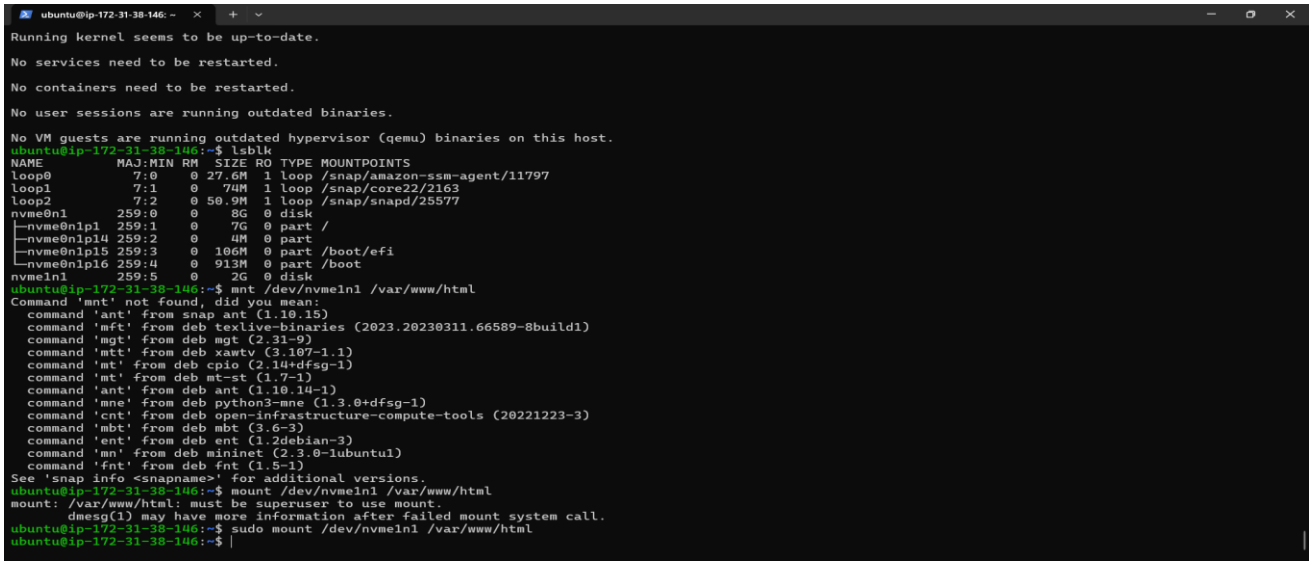


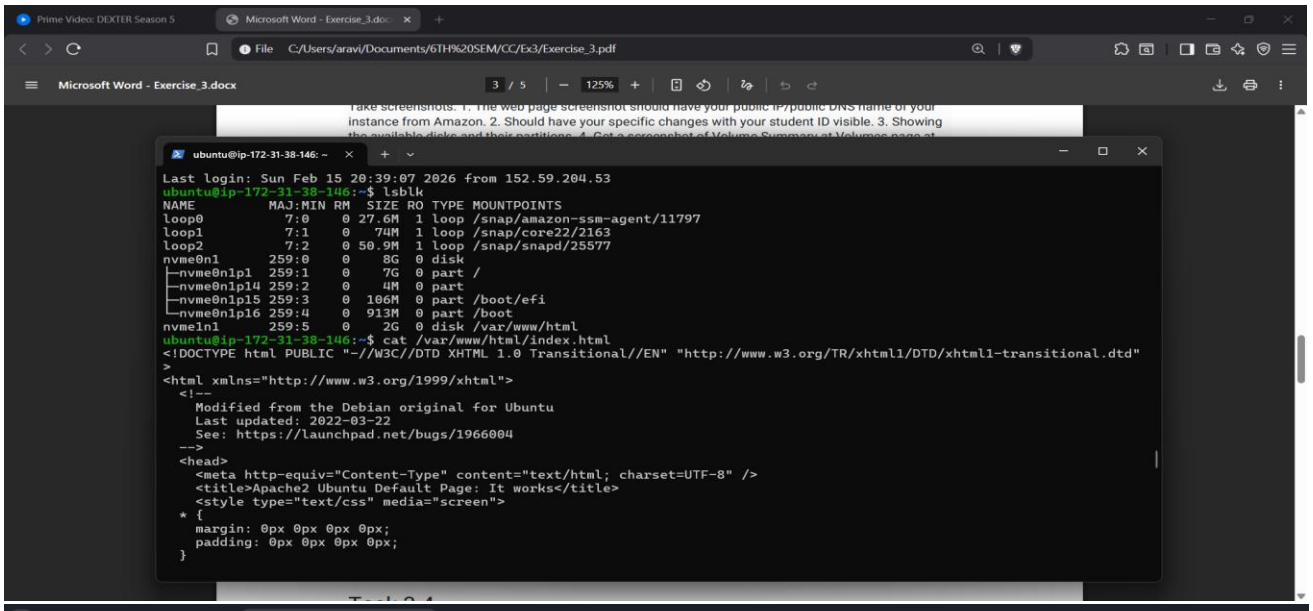
I created another volume for var/www/html which is 23mcce11_vol2 (nvme2n1) I didn't create a partition p1 for it because it is useless and I created /var/www/html using sudo mkdir -p /var/www/html. -p (stands for parents) is to tell mkdir to create parent directories if they don't exist and not to throw error if they exist.

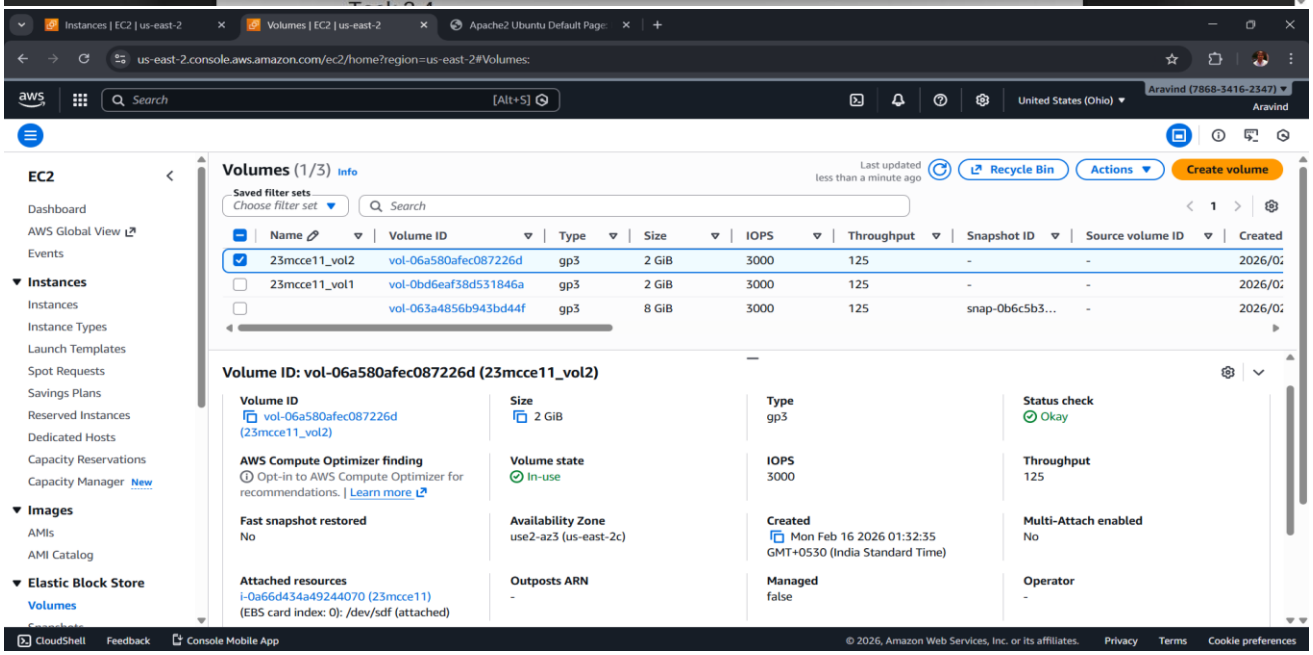


Modification is maddened in "It works!" orange box.

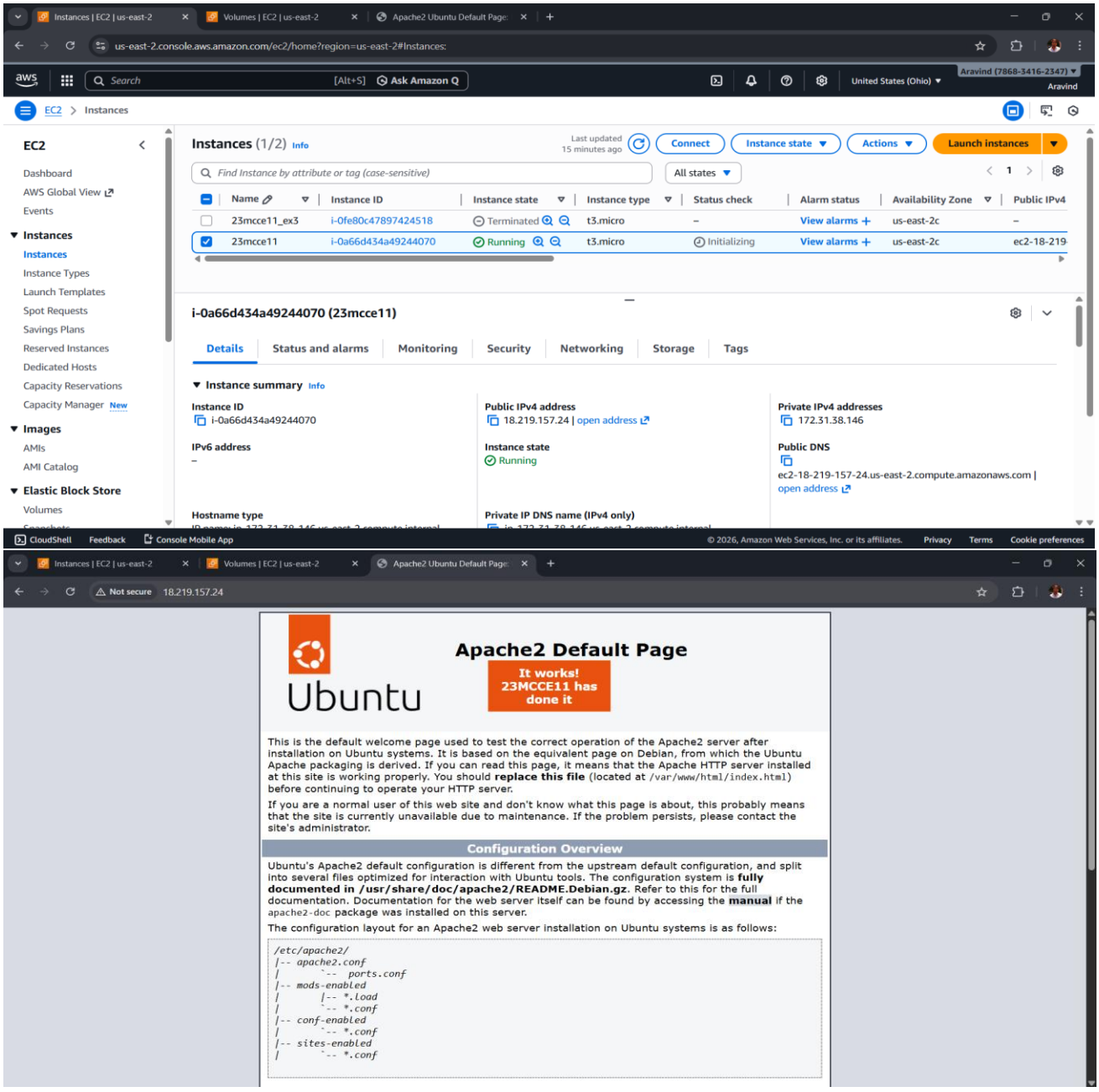
3.


ubuntu@ip-172-31-38-146: ~
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-38-146:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1 7:1 0 74M 1 loop /snap/core22/2163
loop2 7:2 0 50.9M 1 loop /snap/snapd/25577
nvme0n1 259:0 0 8G 0 disk
├─nvme0n1p1 259:1 0 7G 0 part /
├─nvme0n1p14 259:2 0 4M 0 part
├─nvme0n1p15 259:3 0 106M 0 part /boot/efi
└─nvme0n1p16 259:4 0 913M 0 part /boot
nvme1n1 259:5 0 2G 0 disk
ubuntu@ip-172-31-38-146:~$ mnt /dev/nvme1n1 /var/www/html
Command 'mnt' not found, did you mean:
command 'ant' from snap ant (1.10.15)
command 'mft' from deb texlive-binaries (2023.20230311.66589-8build1)
command 'mgt' from deb mgt (2.31-9)
command 'mtt' from deb xawtv (3.107-1.1)
command 'mt' from deb cpio (2.14+dfsg-1)
command 'mt' from deb mt-st (1.7-1)
command 'ant' from deb ant (1.10.14-1)
command 'mne' from deb python3-mne (1.3.0+dfsg-1)
command 'cnt' from deb open-infrastructure-compute-tools (20221223-3)
command 'mbt' from deb mbt (3.6-3)
command 'ent' from deb ent (1.2debian-3)
command 'mn' from deb mininet (2.3.0-1ubuntu1)
command 'fnt' from deb fnt (1.5-1)
See 'snap info <snapname>' for additional versions.
ubuntu@ip-172-31-38-146:~$ mount /dev/nvme1n1 /var/www/html
mount: /var/www/html: must be superuser to use mount.
dmesg(1) may have more information after failed mount system call.
ubuntu@ip-172-31-38-146:~$ sudo mount /dev/nvme1n1 /var/www/html
ubuntu@ip-172-31-38-146:~$


Microsoft Word - Exercise_3.docx
File C:/Users/aravi/Documents/6TH%20SEM/CC/Ex3/Exercise_3.pdf
Microsoft Word - Exercise_3.docx
instance from Amazon. 2. Should have your specific changes with your student ID visible. 3. Showing
ubuntu@ip-172-31-38-146: ~
Last login: Sun Feb 15 20:39:07 2026 from 152.59.204.53
ubuntu@ip-172-31-38-146:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1 7:1 0 74M 1 loop /snap/core22/2163
loop2 7:2 0 50.9M 1 loop /snap/snapd/25577
nvme0n1 259:0 0 8G 0 disk
├─nvme0n1p1 259:1 0 7G 0 part /
├─nvme0n1p14 259:2 0 4M 0 part
├─nvme0n1p15 259:3 0 106M 0 part /boot/efi
└─nvme0n1p16 259:4 0 913M 0 part /boot
nvme1n1 259:5 0 2G 0 disk /var/www/html
ubuntu@ip-172-31-38-146:~$ cat /var/www/html/index.html
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<!--
Modified from the Debian original for Ubuntu
Last updated: 2022-03-22
See: https://launchpad.net/bugs/1966004
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
<title>Apache2 Ubuntu Default Page: It works</title>
<style type="text/css" media="screen">
* {
margin: 0px 0px 0px 0px;
padding: 0px 0px 0px 0px;
}


Instances | EC2 | us-east-2
Volumes | EC2 | us-east-2
Apache2 Ubuntu Default Page:
us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#Volumes:
Search
United States (Ohio)
Aravind
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Volumes (1/3) Info
Last updated less than a minute ago
Recycle Bin
Actions
Create volume
Saved filter sets
Choose filter set
Search
Name
Volume ID
Type
Size
IOPS
Throughput
Snapshot ID
Source volume ID
Created
23mcce11_vol2
vol-06a580afec087226d
gp3
2 GiB
3000
125
2026/02
23mcce11_vol1
vol-0bd6eaf38d531846a
gp3
2 GiB
3000
125
2026/02
vol-063a4856b943bd44f
gp3
8 GiB
3000
125
snap-0b6c5b3...
2026/02
Volume ID: vol-06a580afec087226d (23mcce11_vol2)
Volume ID
vol-06a580afec087226d (23mcce11_vol2)
Size
2 GiB
Type
gp3
Status check
Okay
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. Learn more
Volume state
In-use
IOPS
3000
Throughput
125
Fast snapshot restored
No
Availability Zone
use2-az3 (us-east-2c)
Created
Mon Feb 16 2026 01:32:35 GMT+0530 (India Standard Time)
Multi-Attach enabled
No
Attached resources
i-0a66d434a49244070 (23mcce11)
(EBS card index: 0): /dev/sdf (attached)
Outposts ARN
-
Managed
false
Operator
-
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

I took image 2 to show it is connected to /var/www/html.



4.

I choose Amazon prime video

1. It is used to watch Movies, Series and live tv (new feature but we have to pay extra for every channel). what problem does it solve maybe less cable tv wires hanging on poles.

   Main Features:
   -multiple profiles
   -we can download and watch them in offline (not on web)
   -high quality streaming etc.

2.

2.1) -It does request handling managing thousands may be lakhs of customers at a time

-real time streaming the videos

-video encoding like YouTube

2.2) -It stores video files and also user data and transactions if we rent a movie or buy a channel

- Both, read heavy and write heavy

2.3) People are the users and they are end users and admins will also be there to upload content

2.4) Functional: i) No interruption while watching

ii) control over account and managing subscriptions

Non-Functional: i) 24/7 365 days it should be running, problems may occur

ii) fast networking, maintaining low latency

3.

It needs more elasticity in weekdays and holidays, if any new movies came etc where people binge watch more.

PaaS is more efficient in these times because in PaaS you have to pay for the network you have used that's it no extra cost where in IaaS you have to pay for the instances even, they are partially used.

4.

In Cloud it automatically scales the traffic increases / decreases but in traditional on-premise infrastructure we have to predict and get the hardware in advance to prepare for these.

In cloud we pay as we use how much network (or any other things) but in on-premise we have to buy servers, coolers and have to maintain the servers in cloud we have less maintenance costs.

One of the main is disaster recovery in cloud.

5.

Still chose Amazon Prime video

a) There will be so many that system architect must manage like installing OS's, updating security, installing the files or applications or APIs to handle streaming services, installing Apache, firewall handling and IAM dashboard roles like root, user etc

Compute: EC2

Storage: S3 to store videos

Database: Self-managed / Install MySQL on EC2

Networking: security groups and elastic load balancing etc

Monitoring and scaling: cloud watch and auto scaling

b)

Aws manages servers, OS's, runtime environment, database, monitoring and networking etc

-Developers will have logics for authentication, APIs, database schema, role-based access they focus on what the app does to enhance user experience.

-Amazon Elastic Beanstalk for application deployment and database like Amazon Aurora/DynamoDB for storage same s3 to store videos and monitoring with CloudWatch and AWS X-Ray

c)

Advantages:

i) AWS handles the instances, OS's and scaling operations decreasing operations.

ii) Because of auto scaling it improves networking and faster during traffic spikes.

Disadvantages:

i) No control over system level changes, for high performance it is good to have control and limited access to OS's

ii) AWS PaaS services are designed like so tight or integrated so that it is tough to change the cloud provider even it requires so much redesign to move for on-premise infrastructure.